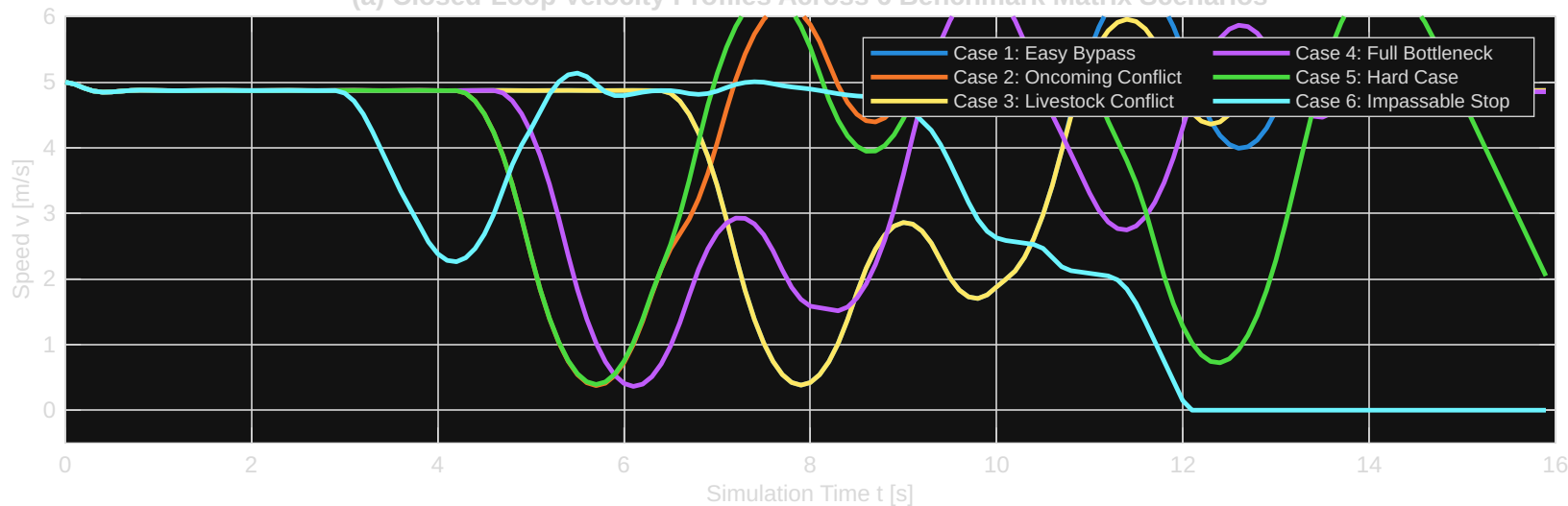


(a) Closed-Loop Velocity Profiles Across 6 Benchmark Matrix Scenarios



(b) Minimum Physical Clearance Margins: Demonstrating Zero Collision Safeguard

